

SEDAR 39 Highly Migratory Species Smoothhound Sharks

Gulf of Mexico Smoothhound Complex

Data Workshop

Life History Working Group Recommendations

1. Identify external characters from genetically verified specimens that will definitively differentiate among the three *Mustelus* species occurring in the northern Gulf of Mexico.
2. Increase tagging effort on the three *Mustelus* species occurring in the northern Gulf of Mexico to gain knowledge pertaining to movement patterns and seasonally mediated distribution.
3. Reexamine all aspects of the species-specific life histories of the three *Mustelus* species occurring in the Gulf of Mexico.
4. Encourage collection of the full suite of body length measurements (i.e. precaudal length, fork length, total length and stretch total length) of all *Mustelus* species occurring in the northern Gulf of Mexico to generate length-length relationships based on a robust sample size.

Commercial Fisheries Working Group Recommendations

1. Given the high difficulty in differentiating among the three species of *Mustelus* occurring in the Gulf of Mexico, even by experienced shark researchers, we feel it is not appropriate to recommend any species-specific identification by fishermen, observers, port samplers, or dealers. Collection of vertebral samples for systematic characterization of age compositions would also require that the whole specimen or a tissue sample be kept for subsequent macroscopic identification or for genetic analysis, respectively.
2. Increase temporal/spatial/fleet-specific shrimp fleet Observer Program coverage to improve bycatch estimates of *Mustelus* species in the shrimp trawl fishery.
3. Conduct research to explore and test the relationship between CPUEs based on shrimp fleet Observer Program and survey (SEAMAP) to indirectly estimate pre-2009 shrimp bycatch CPUE for *Mustelus* species when Observer program data were very limited.

Recreational Fisheries Working Group Recommendations

Given the high difficulty in differentiating among the three species of *Mustelus* occurring in the Gulf of Mexico, even by experienced shark researchers, we feel it is not appropriate to recommend any species-specific identification by fishermen or port samplers. Collection of vertebral samples for systematic characterization of age compositions would also require that the whole specimen or a tissue sample be kept for subsequent macroscopic identification or for genetic analysis, respectively.

Indices of Relative Abundance Working Group Recommendations

- Monitor/record bottom temperature, salinity, DO on all fishery independent surveys

Assessment Workshop

We list below research recommendations that are more feasible and would allow improvement of future stock assessments of this stock:

- Since catches are dominated by shrimp trawl fishery discards, increase the spatio-temporal observer coverage of the shrimp fleet
- Explore the relationship between catch rates derived from the shrimp fleet observer program and those based on the SEAMAP survey to indirectly estimate shrimp bycatch CPUE prior to 2009 when observer program data were especially limited
- Reexamine and/or investigate all aspects of the life histories of the three *Mustelus* species occurring in the Gulf of Mexico

Review Workshop

6. Consider the research recommendations provided by the Data and Assessment workshops and make any additional recommendations or prioritizations warranted.

The panel concurred with the research recommendations of the DW and AW. It is particularly important to maintain the ability to estimate the shrimp trawl bycatch for the future. As more years of data accumulate there will be an improvement in the ability to assess the stock.

The present model software requires that complete catch data are input to the assessment and that they are treated as known error free values. In principle it should be possible to use the shrimp fishery effort data along with the more reliable estimates of catch from 2009 onwards within the model to estimate historical catch and the uncertainties relating to it. It would be desirable to develop such a model which would have wider applicability to stocks that are affected by the same catch data problems.

While it is acknowledged that the species within the smoothhound complex are quite similar biologically, it has been recognised by studies elsewhere (e.g. Gaichas et al. 2012) that individual more vulnerable species within a complex can be adversely affected by aggregated management. This vulnerability may be due to particular species interactions or environmental sensitivity and not just individual species productivity characteristics. Such simulation work could be carried out for the Gulf smoothhound complex to determine whether any of the species may be particularly at risk. The three species in the Gulf smoothhound complex have thus far proved impossible to tell apart visually, and there does not appear to be plans to allow for future estimation of annual total catch per species due to this problem (unless diagnostic morphological features are found). It would be advantageous for future assessments to have such information. Simple and cost effective methods to allow catch estimation per species should be investigated (e.g. random genetic sampling of the catch by observers).

The general SEDAR process is fairly well structured with the development of data workshops and assessment workshops. It is helpful to have the copious documentation. One note is that the rationale for why the decisions were made is often as important as what the decision was. For abundance indices this was often documented in the index worksheets, but not all the decisions were listed in the data workshop report.

Atlantic Smooth Dogfish

Data Workshop

Life History Working Group Recommendations

- Increase tagging effort to examine if there is fine scale structure within *M. canis* off the east coast of the United States to determine if the stock is homogeneous or if it would be more accurately described by northern and southern groupings.
- Conduct genetic analyses in support of Research Recommendation 1.
- Better define seasonal distribution, including regional sex ratios, and identify nursery areas.
- Continue to monitor life history characteristics of *M. canis* off the east coast of the United States to detect potential temporal changes, density-dependent effects or clinal variability among individuals throughout the range.

Commercial Fisheries Working Group Recommendations

- Increase temporal/spatial/fleet-specific shrimp fleet Observer Program coverage to improve bycatch estimates of *Mustelus* species in the shrimp trawl fishery.
- Conduct research to explore and test the relationship between CPUEs based on shrimp fleet Observer Program and survey (SEAMAP) to indirectly estimate shrimp bycatch CPUE for *Mustelus* species when Observer program data were very limited.

Recreational Fisheries Working Group Recommendations

No research recommendations relative to recreational fisheries were formulated.

Indices of Relative Abundance Working Group Recommendations

- Monitor/record bottom temperature, salinity, DO on all fishery independent surveys

Assessment Workshop

1. Modeling considerations.

Improve the fits to length composition data. For example Stock Synthesis allows for the estimation of sex specific selectivity and includes options to utilize parameter offset approaches in the estimation of

selectivity parameters in order to improve parameter estimation. Several methods are also available for selecting among alternative functional forms for selectivity (e.g., Helu et al 2000; Maunder and Harley 2011; Punt et al. 2014). For example, the use of Akaike's information criterion (AIC) (Akaike 1973; Burnham and Anderson 2002; e.g., Hilborn and Mangel 1997) is appropriate for comparing alternative forms of selectivity, as implemented here for comparing proposed base runs Sel-1 and Sel-2, if models compared use the same data and have the same data structure (Helu et al 2000). Alternative methods would be required for selecting among models with different data or with different data structure. For example, the hold-out cross validation has been used for comparison of models run with different data sets (Maunder and Harley 2011).

2. Data Considerations.

Obtain age composition data from existing surveys in order to not have to rely solely only length composition data in the model.

Update age and growth studies in order to resolve potential differences in observed and predicted size at birth.

Review Workshop

TOR 6. Consider the research recommendations provided by the Data and Assessment workshops and make any additional recommendations or prioritizations warranted.

The research recommendations provided by the Data and Assessment Workshops are considered to be reasonable and would strengthen any future assessment. Research recommendations from the data workshop regarding commercial fisheries (point 1.2 in the Research Recommendations Report) focus on developing CPUE and catch estimates from the *Mustelus* bycatch in the Atlantic shrimp fishery. This is a good recommendation because currently the fishery is poorly accounted for in the assessment. However the data workshop indicated that due to the low spatial/temporal overlap (6.5% of the tow hours overlapped with the *M. canis* distribution) there was low probability of significant interaction and inadequate data to develop a catch history. Developing data from this fishery may be impractical given the uncertainty, in addition sampling from the edge of the distribution can be heavily influenced by factors other than abundance, especially in low information situations (SEAMAP recorded only 5 positive tows from 630 over the years 2001 – 2012). A more important research avenue would be to develop better data streams from the gillnet fishery. An additional research recommendation that would assist the SEDAR process from the data standpoint is to increase the monitoring on the gillnet fishery as it is currently the major source of fishing mortality.

Research is required into how to appropriately use the rankings of the CPUE series as weights in the modeling process. An additional need is to conduct research on the estimation of the effective sample

size (appropriate weights) of the length compositions outside the model. Research avenues that would directly assist the stock assessment process are: 1) to consider alternative recruitment functions; 2) using the equivalent of steepness for the Ricker model (as per the Brooks et al. paper) and potentially the low fecundity stock recruitment function which was developed by Ian Taylor for spiny dogfish in the Pacific. Additionally, investigate the modeling of initial depletion in the model (i.e. using estimated fishing mortalities, catches or recruitment offset), and to investigate projections from within SS3. These recommendations are in line with the DW and AW's recommendations of research on how to weight the length composition data, and improving the fits to the selectivities and to obtain age data.

Provide recommendations on possible ways to improve the SEDAR process.

The general SEDAR process is fairly well structured with the development of data workshops and assessment workshops. It is helpful to have the copious documentation. One note is that the rationale for why the decisions were made is often as important as what the decision was. For abundance indices this was often documented in the index worksheets, but not all the decisions were listed in the data workshop report.

Some of the panel's comments regarding the assessment were unable to be addressed because they dealt directly initial model formulation and/or with preliminary analysis. The SEDAR, and in particular the assessment process would have benefited from additional outside input during the assessment workshop (and webinars) from scientists particularly experienced with integrated models such as SS3.